

Rotating Machinery Fault Classification using IWGAN-GP and Small Gray Images

Haidong Shao

College of Mechanical and Vehicle
Engineering
Hunan University
Changsha, China
hdshao@hnu.edu.cn

Wei Li

College of Mechanical and Vehicle
Engineering
Hunan University
Changsha, China
liweili2020@hnu.edu.cn

Min Xia

Yangtze River Delta Research Institute
Northwestern Polytechnical University
Taicang, China
minxia@nwpu.edu.cn

Chengxu Wang

College of Mechanical and Vehicle
Engineering
Hunan University
Changsha, China
wangchengxustudy@163.com

Qinglin Guan

College of Mechanical and Vehicle
Engineering
Hunan University
Changsha, China
tsinglinguan@hnu.edu.cn

Tian'ao Xu

College of Mechanical and Vehicle
Engineering
Hunan University
Changsha, China
xta20000111@163.com

Abstract—Sufficient fault samples of rotating machinery are difficult to be acquired in the application scenarios. The limitation of sample numbers may result in low accuracy and over-fitting in fault classification. Thus, rotating machinery fault classification under small samples is a challenging job. To address the above problem, a fault classification method based on improved Wasserstein generative adversarial network with gradient penalty (IWGAN-GP) is proposed in this paper. Firstly, small two-dimensional (2D) gray images are obtained by converting one-dimensional (1D) raw vibration signals to explore 2D features. Secondly, IWGAN-GP is constructed by using 2D convolution layers to design generator and discriminator to fully extract the features of the 2D gray images. Finally, the generated samples are used for assisting small real samples to achieve fault classification. In order to verify the effectiveness of the proposed method, comparative experiments of bearing fault classification are carried out under different ratios between the generated samples and the real samples. The results show that the proposed method can effectively improve the performance of classifying various fault types of bearing with small samples.

Keywords—rotating machinery, fault classification, IWGAN-GP, small samples, gray images.

I. INTRODUCTION

Rotating machinery is widely used in complex modern industrial equipment, its health monitoring can effectively improve the reliability and reduce the occurrence of casualties. Therefore, how to evaluate the health states and classify different fault types of rotating machinery to make corresponding maintenance strategies has been hot issues [1].

In recent years, due to the strong ability of feature learning and data analysis, fault classification methods based on deep learning have attracted extensive attention of researchers [2-3]. Jia *et al.* [4] proposed a deep neural network structure for intelligent fault diagnosis of rotating machinery, which achieved high accuracy. Jiang *et al.* [5] proposed stacked multilayer noise reduction autoencoder, which could learn the features of the frequency spectrums of the vibration signals. However, these researches are carried out under sufficient fault samples.

The number of samples plays a crucial role in training deep learning models, too few samples will lead to some problems such as over-fitting and low-generalization.

Obviously, it is impossible to let the mechanical equipment run in the failure state for a long time to obtain sufficient fault samples. Therefore, it has great significance to propose an effective method to deal with the fault classification problem with small samples.

The common solutions are to expand the existing samples by generating more samples from the raw samples, such as adding Gaussian noise, signal translation, amplitude drift, and oversampling. These above techniques pay attention to simply transforming the raw samples which may fail to fit the distribution characteristics of small samples. Therefore, the improvement effect of fault classification performance by using these techniques is not significant [6].

Since Goodfellow *et al.* [7] proposed the generative adversarial networks (GAN) in 2014. After GAN has achieved great success in the field of image generation, some researchers utilized the idea of GAN to solve the fault classification problem with small samples. For example, Mao *et al.* [8] combined GAN with stacked denoising autoencoder for generating the frequency signals of raw bearing vibration signals. Wang *et al.* [9] replaced the discriminator network structure with stacked autoencoder and generated spectrum samples by GAN. Luo *et al.* [10] proposed a method which was a combination of deep convolutional GAN and conditional GAN to generate samples of a specified fault type. Li *et al.* [11] added a gradient penalty term to the auxiliary classification GAN (ACGAN), which could generate multiple types of fault samples at the same time.

In these above-mentioned studies, the raw 1D vibration signals are firstly processed and then used as the training samples to input GAN. From the perspective of input's form, the following problems should be considered.

(1) In order to achieve the effective extraction of 1D sample features, it requires that the network structure in the GAN consists of 1D convolution layers or fully connected layers. However, their forward propagation algorithm will lead to large trainable parameters and high complexity.

(2) The trainable parameters of 2D convolution layers are less than 1D convolution layers and fully connected layers. Nevertheless, the feature learning ability may reduce when 2D convolution layers are used to directly extract the features of 1D samples.

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Therefore, it is worthy of designing a strategy to improve GAN so that it can effectively extract the 2D features of 1D vibration signals and realize high-accuracy fault classification under-small samples.

This paper proposes an improved Wasserstein generative adversarial network with gradient penalty (IWGAN-GP) method for rotating machinery fault classification. The main contributions of the proposed method can be summarized as follows. (1) The raw 1D vibration signals are converted to 2D gray images so that 2D features of the raw signals can be adequately explored. (2) Using the 2D convolution layers to design both the generator and discriminator of GAN. In addition, the Wasserstein distance and gradient penalty are used as tricks to stabilize the training process of GAN. (3) Due to the strong feature extraction ability of 2D convolution layers, high-quality 2D gray images can be generated by IWGAN-GP. (4) The generated samples are used for expanding the training samples composed of real samples to achieve fault classification under small real samples.

The remainder of this paper is organized as follows. Section II briefly describes classical GAN, Wasserstein GAN, and Wasserstein GAN with gradient penalty (WGAN-GP). Section III mainly introduces the framework and procedures of the proposed method. Section IV verifies the effectiveness of the proposed method on the bearing dataset. The conclusions and future research works are presented in Section V.

II. BACKGROUND

A. Classical GAN

The classical GAN model proposed by Goodfellow [7] contains two deep neural networks, which are generator and discriminator. As shown in Fig.1, the generator takes samples from random noise distribution and then tries to generate samples whose distribution space is the same as the real samples. The discriminator receives a sample and then predicts whether the input sample is a fake sample (from the generator) or a real sample. The discriminator solves the problem of binary classification, the outputs range of the discriminator is 0 to 1. It should be noted that the training processes of these two networks are alternate, while their parameters are independent.

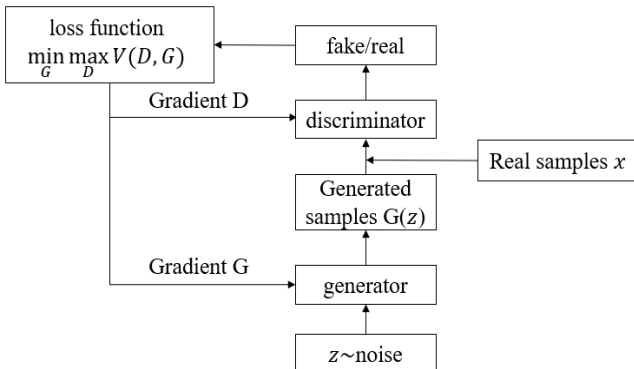


Fig. 1. The structure of classical GAN

Generally, the discriminator and generator are composed of deep neural networks. The objective function of GAN is shown as follows:

$$\min_G \max_D V(G, D) = E_{x \sim P_{data(x)}} [\log(D(x))] + E_{z \sim P_{z(z)}} [\log(1 - D(G(z)))] \quad (1)$$

where $V(G, D)$ is the objective function, G and D represent generator and discriminator, respectively, x is collected from $P_{data(x)}$ which represents the distribution of real samples, z is the noise randomly sampled from normal distribution, $G(z)$ is a sample generated by the generator, $P_{z(z)}$ represents the distribution of the generated samples, and $D(x)$ means the outputs of discriminator when the input is real samples. The objective function is called JS (Jensen-Shannon) divergence, which needs to be optimized in the following two parts during training.

(1) **Discriminator training.** $E_{x \sim P_{data(x)}} [\log(D(x))]$ and $E_{z \sim P_{z(z)}} [\log(1 - D(G(z)))]$ need to be maximized, meaning the discriminator aims to correctly distinguish the inputs, so that the objective function wants the output $D(x)$ to get close to 1 and $D(G(z))$ to 0.

(2) **Generator training.** The parameters of the discriminator should be frozen at first, and then minimize $E_{z \sim P_{z(z)}} [\log(1 - D(G(z)))]$, meaning that the discriminator cannot distinguish the generated sample as fake.

The training process is the minimax game between generator and discriminator. After a number of iterations, the discriminator cannot distinguish between real samples and fake samples, and the final goal of GAN to generate high-quality samples is achieved.

However, due to the discreteness of JS divergence, the classical GAN has the following problems in the training process. (1) Model collapse, although generated samples are similar to real samples, the diversity of generated samples is not enough. (2) Gradient vanishing, in the training process, it is difficult to achieve the convergence of GAN models, and the gradient vanishing will result in the stopping of the weights update.

B. Wasserstein GAN

Wasserstein GAN [12] (WGAN) replaced JS divergence with Wasserstein distance (also known as EM distance), and imposed the Lipschitz condition on the discriminator to constrain the gradient. The definition of Wasserstein distance is shown as follows:

$$W(p, q) = \inf_{g \sim \Pi(p, q)} E_{(x, y)} [\|x - y\|] \quad (2)$$

where W represents the Wasserstein distance between two distributions, $\inf_{g \sim \Pi(p, q)}$ is the set of all possible joint distributions of real samples p and generated samples q , x and y represent one of real samples and generated samples, respectively, and $E_{(x, y) \sim g} [\|x - y\|]$ is the expectation value of the distance between p and q .

In brief, WGAN has made three changes compared with the classical GAN. (1) The sigmoid function is removed from the last layer of the discriminator. (2) The objective functions of the discriminator and generator do not carry out logarithmic operations. (3) After updating the parameters of the discriminator each time, the absolute values of weight parameters are restricted to no more than a fixed constant

(satisfying the Lipschitz constraint condition). These improvements effectively eliminate the instability of GAN training.

C. Wasserstein GAN with Gradient Penalty

Although WGAN analyzes the reasons for the instability of the GAN training process from the perspective of the objective function, and proposes effective solutions, the implementation of weight clipping in WGAN still leads to two problems. (1) The way of applying Lipschitz constraint to the discriminator is unreasonable. Most of the weights of the network converge at the two ends of the constraint interval, which makes the learning process of the discriminator become simple function mapping. (2) The forced clipping of the network weight may cause the gradient vanishing problem or the gradient exploding problem.

Wasserstein GAN with Gradient Penalty (WGAN-GP) [13] solved the gradient vanishing problem and gradient exploding problem. Gradient clipping is replaced by adding gradient penalty to the objective function. The gradient penalty can constrain the discriminator gradient value below 1, which makes the discriminator still meet the Lipschitz constraint condition. The definition of gradient penalty and the objective function of WGAN-GP are shown as equations (3) and (4) :

$$GP = E_{\hat{x} \sim P_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad (3)$$

$$LOSS = E_{x_r \sim P_{data}} [D(x_r)] - E_{x_f \sim P_{z(z)}} [D(x_f)] - \lambda * GP \quad (4)$$

where $\hat{x}$ is defined by sampling uniformly along straight lines between the distribution P_{data} of real samples and the distribution $P_{z(z)}$ of generated samples, $\nabla_{\hat{x}} D(\hat{x})$ is the gradient of the discriminator, and λ is the coefficient of gradient penalty.

III. FRAMEWORK AND PROCEDURES OF THE PROPOSED METHOD

A. Signal-to-image Conversion

In order to let the 2D GAN can effectively extract the features of 1D vibration signals. This paper considers converting the 1D vibration signals to 2D gray images, and then uses the 2D gray images as the inputs. Here, the technique of signal-to-image conversion proposed in [14] is used, which provides a way to explore 2D features of raw vibration signals. The conversion algorithm is shown as follows:

$$P(j, k) = \text{round} \left\{ \frac{L((j - 1) * M + k) - \min(L)}{\max(L) - \min(L)} * 255 \right\} \quad (5)$$

where $P(j, k)$ represents the value of row j and column k in a 2D gray image, M represents the side length of the image matrix with size of $M * M$, L is a randomly selected 1D signal whose length is $M * M$, and round function is used to normalize the value of a pixel to an integer between 0 and 255.

B. IWGAN-GP construction

In this study, IWGAN-GP is mainly constructed by 2D convolution layers rather than 1D convolution layers or fully connected layers. In addition, Wasserstein distance and gradient penalty are used for stabilizing the training process. The calculation process [2] of the 2D convolution layers is shown as follows:

$$x_{i,j}^c = \sum_{h=1}^H \sum_{l=1}^L k_{h,l}^c * x_{i+h-1,j+l-1}^{c-1} \quad (6)$$

where $x_{i,j}^c$ is the value of row i and column j in the c th layer, $x_{i+h-1,j+l-1}^{c-1}$ is the value of row $(i + h - 1)$ and column $(j + l - 1)$ in the $(c - 1)$ th layer, $k_{h,l}^c$ is the h, l th weight of kernel in the c th layer, H and L are the height and length and depth both of the kernels.

The structures and hyperparameters selection of both generator and discriminator are crucial for the overall model. After a long time of adjustment, the details of IWGAN-GP are adopted as follows.

(1) The generator is designed with the up-sampling operations (Conv2Dtranspose layers) which can transform the input random noise into a 2D image by matrix operations. The discriminator is designed by improving the structure of the LeNet.

(2) Batch normalization (BN) operation is added into the generator for preventing the gradient vanishing and avoiding over-fitting. It should be mentioned that the BN operation cannot be added into the discriminator due to that the gradient penalty has already been used in the discriminator.

(3) In the generator, the Tanh function is used as the activation function of the output layer, and the Relu function is used in rest of layers. In the discriminator, the Leaky Relu function is used as the activation function, it should be noted that the output layer does not use activation functions.

(4) In order to avoid excessively strong learning ability of the discriminator which may result in the generator cannot generate high-quality samples. The discriminator is trained five times and then the generator is trained once in one iteration. Adam is selected as the algorithm of optimizer. The details of IWGAN-GP are shown in Table I.

TABLE I. DETAILS OF IWGAN-GP

Networks	Descriptions	Operations	Activation Functions
Discriminator	Conv2D (5*5*32)	None	Leaky Relu
	Conv2D (3*3*64)	None	Leaky Relu
	Conv2D (3*3*128)	None	Leaky Relu
	Conv2D (3*3*256)	None	Leaky Relu
	Dense (256)	None	Leaky Relu
	Dense (128)	None	Leaky Relu
Generator	Dense (1)	None	None
	Dense (8192)	Reshape (None,4,4,512)	None
	Conv2Dtranspose (5*5*128)	BN	Relu
	Conv2Dtranspose (5*5*64)	BN	Relu
	Conv2Dtranspose (5*5*32)	BN	Relu
	Conv2Dtranspose (5*5*1)	BN	Tanh

C. The Overall Framework

The overall framework of the proposed fault classification method is shown in Fig.2. The procedures are provided as follows: Firstly, the small 1D vibration signals are converted into 2D gray images as the final training samples. Secondly, the 2D discriminator and 2D generator of IWGAN-GP are trained with alternate optimization. When the Nash equilibrium is reached, more 2D gray images are generated to expand the small real samples. Finally, the expanded samples are input to a classifier built with 2D convolutional neural network (2D CNN). Test samples entirely composed of real samples are input into the trained 2D CNN and output the classification results.

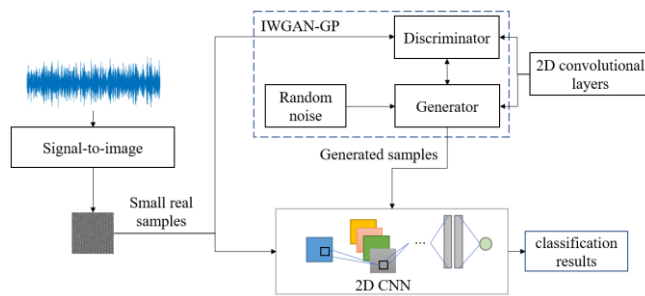


Fig. 2. Overall framework of the proposed method

IV. EXPERIMENTS AND RESULTS DISCUSSION

Bearing is one of the most important parts of rotating machinery, in order to evaluate the performance of the proposed method, several groups of experiments are carried out on Case Western Reserve University (CWRU) bearing dataset.

A. CWRU Bearing Dataset

The CWRU dataset is collected by the acceleration sensors placed in the fan end (FE) whose sampling frequency is 12 kHz and placed in the drive end (DE) whose sampling frequencies are 12 and 48 kHz. According to the location of the damage, it can be divided into three types: rolling element (RE), inner ring (IR) and outer ring (OR) with the four sizes of damage diameters such as 0.18 mm, 0.36 mm, 0.54 mm and 0.72 mm. In addition, there are four different loads (0 HP to 3 HP) of working conditions.

In this study, DE samples were obtained with 12 kHz under the working load of 0 HP. According to the damage sizes of 0.18 mm, 0.36 mm and 0.54 mm, one normal condition and nine fault conditions are selected. 4096 sampling points of 1D vibration signal are used as a sample and converted to an image whose size is 64*64 using conversion technique. The converted images are shown in Fig. 3, it can be observed that gray images under different conditions look different from each other.

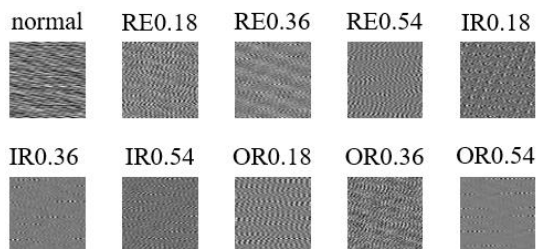


Fig. 3. Converted gray images of the 10 conditions

B. Influence of Small Samples on Fault Classification

In order to study the influence of small samples on fault classification, the 2D CNN fault classification models were trained by different limited numbers of training samples. It should be mentioned that the details such as structures and parameters were not changed in all experiments. The details of the 2D CNN fault classification model are shown in Table II.

TABLE II. DETAILS OF 2D CNN CLASSIFICATION MODEL

Descriptions	Operations	Activation Functions
Conv2D (5*5*32)	Maxpool and BN	Relu
Conv2D (3*3*64)	Maxpool and BN	Relu
Conv2D (3*3*128)	Maxpool and BN	Relu
Conv2D (3*3*256)	Maxpool and BN	Relu
Dense (256)	None	Relu
Dense (128)	None	Relu
Dense (10)	None	Softmax

Table III shows the results of comparison experiments, and the each accuracy rate is the average classification accuracy of each experiment repeated ten times. From the results, it is not difficult to draw a conclusion that the small samples will lead to low accuracies of fault classification. Therefore, it is very necessary to obtain sufficient training samples for achieving relatively high accuracies.

TABLE III. RESULTS OF COMPARISON EXPERIMENTS

Training samples	Testing samples	Accuracies /%
4000 (400*10)	1000 (100*10)	99.8
2500 (250*10)	1000 (100*10)	99.1
2000 (200*10)	1000 (100*10)	97.8
1500 (150*10)	1000 (100*10)	96.6
1000 (100*10)	1000 (100*10)	91.9
750 (75*10)	1000 (100*10)	87.2
500 (50*10)	1000 (100*10)	78.7
250 (25*10)	1000 (100*10)	54.0

C. The Generated 2D Gray Images of IWGAN-GP

In this part, the evolution of generated samples and loss functions during the training are presented to explain the IWGAN-GP how to generate samples.

It should be noted that 250 (25*10) real samples were selected as the training samples of IWGAN-GP. Finally, 1000 samples for each condition (a total of 10000) were generated to expand the small real samples.

Fig.4 shows the curves of the loss functions for the generator and discriminator in the training. It can be seen that the fluctuating range of loss function of the discriminator is sharp at the beginning of training. After some iterations the loss function is convergent, indicating that the training process is effective. After about 6000 iterations, the fluctuating range of two-loss functions is kept in a small range. It shows that the adversarial training process between the generator and discriminator is in Nash equilibrium.

Fig.5 illustrates the evolution of the generated 2D gray image samples during training. The discriminator learns to distinguish between the generated samples and real samples. The generator tries to cheat the discriminator which is more

and more powerful, and the quality of generated samples becomes higher after some iterations.

Fig. 6 shows the 10 real samples and their corresponding final generated samples by IWGAN-GP. Due to that inputs of the generator are random noises of normal distribution, there exist slight differences among the generated samples and real samples.

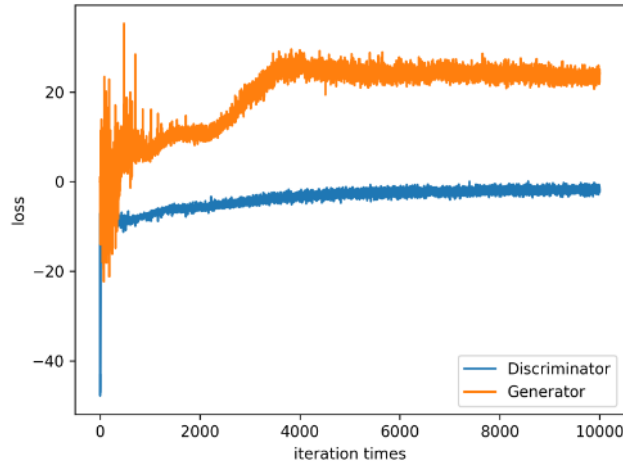


Fig. 4. Loss function values during training

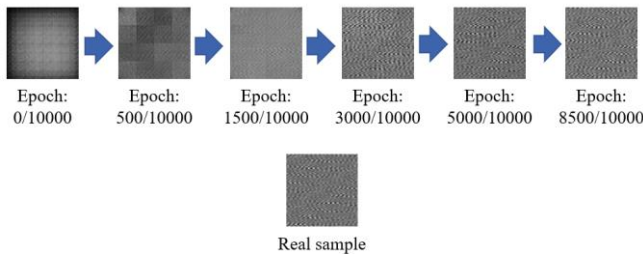


Fig. 5. Evolution of the generated samples in training

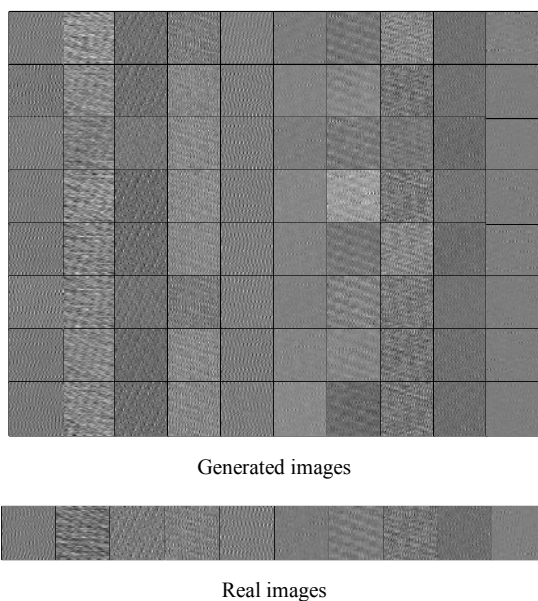


Fig. 6. Comparison between the real samples and generated samples

D. Fault Classification Using the Real Samples and Generated Samples

In this study, the generated samples are input into the 2D CNN fault classification model to verify the performance of the proposed method.

The strategy for validation is using different training samples to determine whether the generated samples improve the performance of classification. There are three different kinds of training samples for comparative experiments. (1) The training samples only include real samples (Experiments 1-7). (2) The training samples include both the generated samples and real samples (Experiments 8-15). (3) The training samples only include generated samples (Experiments 16-18).

The results of comparative experiments are shown in Table IV, and the accuracy rate is the average classification accuracy of each experiment repeated ten times. From the comparative results of Experiment 3 to 7 and Experiment 8 to 15, the performance of fault classification can be improved effectively by using generated samples to expand the training samples.

However, from the comparative results of Experiment 4 to 6 and Experiment 12 to 15. The accuracies of training samples which including generated samples and real samples are lower than the same number training samples which are composed of all real samples. In addition, combined with the results of Experiment 16 to 18. It can conclude that there are still some differences between the generated samples and the real samples.

TABLE IV. COMPARATIVE RESULTS

Experiments	Real samples	Generated samples	Accuracies /%
1	4000 (400*10)	0	99.8
2	2500 (250*10)	0	99.1
3	2000 (200*10)	0	97.8
4	1500 (150*10)	0	96.6
5	1000 (100*10)	0	91.9
6	500 (50*10)	0	78.7
7	250 (25*10)	0	54.0
8	2000 (200*10)	2000 (200*10)	99.5
9	2000 (200*10)	500 (50*10)	98.5
10	1000 (100*10)	1500 (150*10)	98.2
11	1000 (100*10)	1000 (100*10)	97.2
12	500 (50*10)	1000 (100*10)	93.2
13	500 (50*10)	500 (50*10)	89.1
14	250 (25*10)	750 (75*10)	80.9
15	250 (25*10)	250 (25*10)	69.8
16	0	2000 (200*10)	89.8
17	0	1000 (100*10)	84.2
18	0	500 (50*10)	68.9

V. CONCLUSION

A new fault classification method based on IWGAN-GP is proposed in this paper. The IWGAN-GP are designed by ideas of 2D convolution layers and WGAN-GP, which can generate samples that are similar to real samples by learning the features of the 2D gray images. The generated samples are added to the training samples for improving the accuracy of fault classification under small samples.

CWRU bearing dataset is used to validate the effectiveness of the proposed method, and the results show that the proposed method can help improve fault classification accuracy with small real samples. However, the proposed IWGAN-GP only can generate one type of fault samples at one time. Referring to the idea of ACGAN, we will design a novel GAN which can learn the features of samples under various conditions at the same time and generate corresponding samples.

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